

```
CREATE DATABASE SchoolDB;
USE SchoolDB;
```

```
CREATE TABLE Teachers (
    teacher_id INT PRIMARY KEY,
    teacher_name VARCHAR(50),
    email VARCHAR(100)
);
```

```
CREATE TABLE Subjects (
    subject_id INT PRIMARY KEY,
    subject_name VARCHAR(50),
    teacher_id INT,
    FOREIGN KEY (teacher_id) REFERENCES Teachers(teacher_id)
);
```

```
CREATE TABLE Students (
    student_id INT PRIMARY KEY,
    student_name VARCHAR(50),
    grade INT,
    city VARCHAR(50)
);
```

```
CREATE TABLE Exam_Marks (
    record_id INT PRIMARY KEY,
    student_id INT,
    subject_id INT,
    marks INT,
    exam_date DATE,
    FOREIGN KEY (student_id) REFERENCES Students(student_id),
    FOREIGN KEY (subject_id) REFERENCES Subjects(subject_id)
);
```

```
-----

INSERT INTO Teachers VALUES
(1, 'Mr. Perera', 'perera@school.lk'),
(2, 'Mrs. Silva', 'silva@school.lk'),
(3, 'Mr. Gunawardena', 'guna@school.lk');
```

```
INSERT INTO Subjects VALUES
(101, 'Mathematics', 1),
(102, 'Science', 2),
(103, 'History', 3),
(104, 'English', 1);
```

```
INSERT INTO Students VALUES
(1, 'Kasun Kalhara', 10, 'Colombo'),
(2, 'Amal Perera', 10, 'Kandy'),
(3, 'Nimali Fonseka', 11, 'Galle'),
(4, 'Ruwan Hettiarachchi', 11, 'Colombo'),
(5, 'Saman Kumara', 10, 'Matara');
```

```
INSERT INTO Exam_Marks VALUES
(1, 1, 101, 85, '2023-12-01'), -- Kasun - Maths
(2, 1, 102, 60, '2023-12-02'), -- Kasun - Science
(3, 2, 101, 45, '2023-12-01'), -- Amal - Maths
```

(4, 3, 103, 92, '2023-12-03'), -- Nimali - History
(5, 4, 101, 78, '2023-12-01'), -- Ruwan - Maths
(6, 5, 102, 30, '2023-12-02'); -- Saman - Science

Quactions

1) Get a list of all the names of students in the school and their cities.

```
select student_name,city from Students;
```

2) Get details of students living in the city of 'Colombo'.

```
select * from Students  
where city = 'Colombo';
```

3) Obtain the Mark Records of students who scored more than 75 marks.

```
select * from Exam_Marks  
where marks >= 75;
```

4) How many students are studying in Grade 10?

```
select COUNT(*) from Students where grade = 10;
```

5) What is the average score obtained by students in Mathematics (Subject ID = 101)?

```
select AVG(marks) from Exam_Marks  
where subject_id = 101;
```

6) Sort the exam scores from highest to lowest.

```
select * from Exam_marks order by marks desc;
```

7) Who are the students whose names start with the letter "K"?

```
select * from Students where student_name like 'K%';
```

8) Get the reports of students who appeared for Science (Subject ID = 102) but scored less than 40 marks.

```
select * from Exam_marks where subject_id = 102 AND marks<40;
```

9) What is the number of subjects currently taught at the school?

10) Show cases where scores were between 50 and 80 (inclusive).

```
select * from Exam_marks where marks between 50 AND 80;
```

11) Search only for Student Names who appeared for the subject "Mathematics" (Subject ID 101).

12) Which students scored higher than the class average?

```
select * from Exam_marks
```

```
where marks > (select AVG(marks) from Exam_marks);
```

13) How many students appeared for each subject (Subject ID)?

```
select subject_id,  
COUNT(student_id) as student_count  
from Exam_marks  
group by subject_id;
```

14) What is the total marks obtained by each student (Student ID)?

```
select student_id,  
SUM(marks) as Total_marks  
from Exam_Marks  
group by student_id;
```

15) What are the IDs of students who wrote exams in more than one subject (Subject count > 1)?

```
select student_id,  
COUNT(subject_id) as exam_count  
from Exam_marks  
group by student_id  
HAVING COUNT(subject_id)>1;
```

16) Find the names of students who did not appear for the exam (not in the Exam_Marks table).

17) Get the score details of the student who scored the highest mark.

```
select * from Exam_marks  
where marks = (select MAX(marks) from Exam_marks);
```

18) Show whether students "Pass" or "Fail" based on their scores (Pass if scores >= 50, Fail otherwise).

```
select student_id,subject_id,marks,  
case  
    when marks >= 50 then 'PASS'  
    else 'FAIL'  
end  
as Exam_result  
from Exam_marks;
```

19) Get the subject names from Z to A, not in alphabetical order.

```
select subject_name
from subjects
order by subject_name desc;
```

20) Show only the Top 3 highest scoring records.

```
select*from Exam_marks
order by marks desc
LIMIT 3;
```

Joins

Level 1: Easy

1) Show the names of the students and the marks they obtained.

```
select Students.student_name, Exam_Marks.marks
from Students
inner join Exam_marks
on Students.student_id = Exam_marks.student_id;
```

2) Show the name of each subject (Subject Name) and the name of the teacher who teaches it (Teacher Name).

```
select Subjects.subject_name, Teachers.teacher_name
from Subjects
inner join Teachers
on Subjects.teacher_id = Teachers.teacher_id;
```

3) What are the names of the students and the dates they sat for the exam?

```
select Students.student_name, Exam_marks.exam_date
from students
inner join Exam_marks
on Students.student_id = Exam_marks.student_id;
```

4) Show only marks related to Mathematics (with Subject Name).

```
select Subjects.subject_name, Exam_marks.marks
```

```
from Subjects
inner join Exam_marks
on Subjects.subject_id = Exam_marks.subject_id
where Subjects.subject_name = 'Mathematics';
```

5) Show the name of each subject and the email address of the corresponding teacher.

Level 2: Medium

1) Show all three: the student's name, the subject name, and the marks obtained.

```
select
    Students.student_name,
    Subjects.subject_name,
    Exam_marks.Marks
from Exam_marks
join Students on Exam_marks.student_id =
Students.student_id
join Subjects on Exam_marks.subject_id =
Subjects.subject_id;
```

2) What subjects did student 'Kasun Kalhara' appear for and who are the teachers who teach them?

```
select
    Subjects.subject_name,
    Teachers.teacher_name
from Exam_marks
join Students on Exam_marks.student_id =
Students.student_id
join Subjects on Exam_marks.subject_id =
Subjects.subject_id
join Teachers on Subjects.teacher_id =
Teachers.teacher_id
where Students.student_name = 'Kasun Kalhara';
```

3) Show the marks obtained by students in the city of 'Colombo' and the relevant subjects.

```
select
```

```
Students.student_name,  
Subjects.subject_name,  
Exam_marks.marks  
from Students  
join Exam_marks on Students.student_id =  
Exam_marks.student_id  
join Subjects on Exam_marks.subject_id =  
Subjects.subject_id  
where Students.city = 'Colombo';
```

4) What are the names and subjects of the students who scored more than 80?

```
select  
    Students.student_name,  
    Subjects.subject_name,  
    Exam_marks.marks  
from Exam_marks  
join Students on Exam_marks.student_id =  
Students.student_id  
join Subjects on Exam_marks.subject_id =  
Subjects.subject_id  
where Exam_marks.marks > 80;
```

5) Who were the students who appeared for the subjects taught by "Mr. Perera"?

```
select  
    Students.student_name,  
    Subjects.subject_name  
from Teachers  
join Subjects on Teachers.teacher_id =  
Subjects.teacher_id  
join Exam_marks on Subjects.subject_id =  
Exam_marks.subject_id  
join Students on Exam_marks.student_id =  
Students.student_id
```

```
where Teachers.teacher_name = 'Mr. Perera';
```